Reproducing Laya's numbers before beating them: a failed full-split reproduction, a cleared harness, and evidence that option wording, more than the option budget, drives many-option failure

PinoAI research note · THEMIS-BENCH-001 · v1.6, full split and confirmatory 0.3.20 run in

Stan Lee V. Natividad
Kooya PH AI Research
stan@kooya.ph

2026-10-07

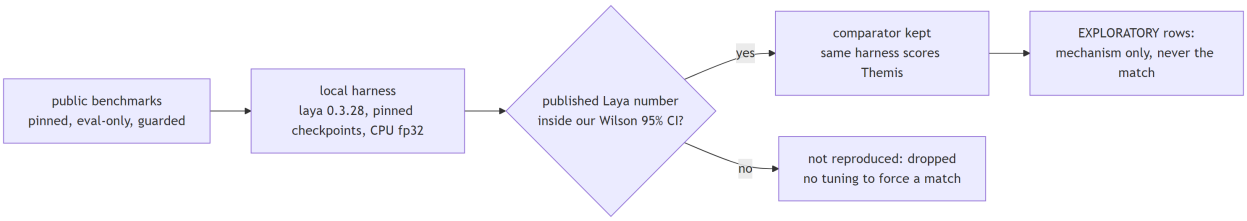
Abstract

A comparison is only as good as the harness that scores both sides, so the first Themis experiment scored nothing of ours: it asked whether our local harness, running Laya's pinned checkpoints on public benchmarks, reproduces the numbers Laya's authors published. Two results and one finding.

1. **AG News: pilot slice matches exactly; pre-declared full-split criterion NOT met.** 0.95, Wilson 95% [0.888, 0.979], on the 100-item BTZSC pilot manifest, with the published 0.950 inside. On the full 7,600-item test split - the criterion declared before any run - Laya 0.3.28 scores 7,019/7,600 = 0.9236 [0.9174, 0.9293], 0 errors, and 0.950 lies outside. By the pre-declared rule this is recorded as "not reproduced"; no document re-words it as a match. The harness is nonetheless CLEARED for paired use (section 3a): it is slice-consistent item by item, and the leading reading is that the authors' 0.950 is a ~100-item estimate (unverified: n unstated).
2. **Banking77: consistent, not reproduced.** The single pre-registered reproduction configuration (chosen before the run: 77 PolyAI category names rendered with spaces, default head budget, default routing) scores 0.37 [0.28, 0.47]. The published 0.425 lies inside the interval, so by the pre-declared rule Banking77 stays a comparator - but the screen is weak at n = 100 and our point estimate is below the published figure. The paper says "consistent within a weak screen", never "reproduced 0.425".
3. **The finding (exploratory, n = 100, one run per row): wording, more than budget.** On the same 100 items, BTZSC's 72 sentence-style options ("This example is …") give 0.02 under both routings - chance for 72 options is 1/72 = 0.014. Raising the option budget from 256 to 1,024 tokens gives 0.09 [0.048, 0.162] on the forced multilingual route (the English default errors out): a small but significant paired gain (7 vs 0 discordant, McNemar exact p = 0.016). Switching to the 77 PolyAI category names gives 0.37 [0.282, 0.468]: decisive, paired (37 vs 2 discordant, p = 2.8e-09), and the two intervals do not overlap, so a 4x budget does not close the gap. A "shared prefix stripped" row (0.01/0.02; 0.07 with the 1,024 budget) turned out ineffective: 71 of the 72 BTZSC options open "This banking customer example message is …", one opens "This customer example message is …", so the common-prefix strip removed one word. The shared-prefix hypothesis is therefore UNTESTED. Forced multilingual on the BTZSC options collapses onto option 0 for 98 of 100 items - the first position AND the only option with a different opening - so position bias and opening salience are not separated. One confound must be stated: the names row changes the option WORDING and the option SET together (72 BTZSC hypotheses with BTZSC gold vs 77 PolyAI categories with PolyAI gold; every row is scored in its own label space, and no item lacks a correct option). The direction still favours wording - more options, yet far higher accuracy - but this is a wording/label-set contrast, not a one-variable ablation.

Consequences for Themis: it is scored against Laya on identical option text and framing (otherwise a "win" is a framing artefact), Laya always runs at its best released configuration, and Themis's option-wise scoring is motivated by the wording result.

1. Why reproduce first



No Themis number before the harness reproduces Laya's own numbers.

Rules fixed before any run (ledger 7fd27f, 1c99a3, 3ac327): one reproduction configuration per benchmark, declared in advance; the match test is "published number inside our Wilson 95% interval"; an n = 100 pass is labelled a weak screen; a miss means "not reproduced" and the benchmark is dropped as a comparator rather than tuned; every other configuration is an exploratory mechanism row that can never become the match. The commercial judge is not run: its numbers are cited as third-party published, unverified and not paired.

2. Setup

Laya 0.3.28 at the registered pins (laya 55cf4c4e, laya-multilingual e4e9ddf2, laya-typed-decisions 1a793eb5), CPU, fp32 (no autocast), both routings (Laya's default Router and forced multilingual). Items: the BTZSC pilot manifests (100 deterministic, roughly class-balanced test items per dataset). Gold labels: the BTZSC rows are scored against BTZSC's own gold over its 72 hypotheses; the pre-registered Banking77 row maps the same 100 items by exact text to the original PolyAI test set (100/100 matched) and scores against the PolyAI category among 77. DAIR Emotion is cited only, never run (its licence is research and educational use only; governance ruling 0f836c).

3. Results

Table 1: Harness check: pilot slice (n = 100, weak screen) and full split.

Benchmark	Configuration	Routing	Accuracy	Wilson 95%	Published (Laya)	Status
AG News	BTZSC pilot, 4 labels	default	0.95	[0.888, 0.979]	0.950	pilot slice: inside
AG News	FULL test split, 7,600, PRE-DECLARED criterion	default	0.9236	[0.9174, 0.9293]	0.950	NOT met: 0.950 outside; recorded as not reproduced
AG News	FULL test split, 7,600	forced multilingual	0.9089	[0.9023, 0.9152]	-	reported; less accurate, better calibrated (ECE 0.032 vs 0.056)
AG News	same	forced multilingual	0.94	[0.875, 0.972]	-	reported
Banking77	PRE-REGISTERED: 77 PolyAI names, spaces, default budget	default	0.37	[0.282, 0.468]	0.425	consistent within a weak n = 100 screen
Banking77	BTZSC sentence options (72)	default / multilingual	0.02 / 0.02	[0.006, 0.070]	-	framing row

Table 2: Mechanism rows on the same 100 Banking77 items, one run each.

Exploratory row (never the match), 72 BTZSC options, chance 0.014	default	forced multilingual (Wilson 95%)
BTZSC options, "shared prefix stripped" - INEFFECTIVE (1 word removed; prefix hypothesis untested)	0.01	0.02 [0.006, 0.070]
BTZSC options, head budget 256 -> 1,024	100 runtime errors (no score)	0.09 [0.048, 0.162]
both	100 runtime errors (no score)	0.07 [0.034, 0.138]

Table 3: Paired tests (ml-engineer 962f3e; the first two recomputed independently by research from the preds files and asserted by the checker).

Paired contrast (same 100 items)	only first correct	only second correct	McNemar exact p
77 PolyAI names vs 72 BTZSC sentences, default routing	37	2	2.8e-09
budget 1,024 vs 256, forced multilingual, BTZSC options	7	0	0.016
77 PolyAI names vs BTZSC strip + 1,024, multilingual	32	2	6.9e-08

Reading the rows: the BTZSC framing sits at chance (0.02 vs 1/72). A 4x budget (14 tokens per option at 72) helps a little and significantly (0.02 -> 0.09, p = 0.016) but its interval does not reach the names row's [0.282, 0.468]. Changing to the 77 category names moves the same items to 0.37 (p = 2.8e-09) - but that row also changes the label set and gold (section 2), so the contrast is wording/label-set, not wording alone; every row is scored in its own label space and no item lacks a correct option, so there is no accuracy cap hiding in the BTZSC rows. The strip rows did not test the shared-prefix hypothesis: the strip removed only "This", because one of the 72 options opens differently (examples - BTZSC: "This banking customer example message is about activating a card."; PolyAI name: "card linking"). What the ineffective strip hid: forced multilingual collapses onto option 0 (first position, and the only option with a different opening) for 98 of 100 items; cause not separated, since position-0 bias and opening salience predict the same count - and with only "This" stripped, option 0 still opens differently yet the collapse drops to 18/100, so neither explanation fits cleanly yet. The registered follow-up (exploratory, after the full AG News slice, no family-wise claim) is three rows: a correct majority-prefix strip with a no-shared-trigram assert; a POSITION control that moves the differing-opening option to index 36 with gold remapped; and all options normalised to one opening. Each asserts the same 100 items and option multiset by sha256. The default-routing 1,024-budget rows errored on every item; the preds files carry the error text ("only 37 of its 72 option markers fit in max_len=512 with head_max_len=1024"), so the cause is the English checkpoint's `max_len` of 512 below the requested option budget (THEMIS-ENG-001). (Laya's loader also warns that its 11+-option temperature is clamped and those confidences uncalibrated - a second reason many-option decisions are Laya's weak regime.)

3a. The failed criterion, and why the harness is still cleared

The pre-declared test (ledger 7fd27f / 1c99a3) was "published number inside our full-split Wilson 95% interval; a miss means not reproduced, dropped as a reproduction target, never tuned". It failed, and the paper says so. What the test did not separate - a registration error of research's, recorded here - is "our harness is correct" from "the authors' unpaired figure is reproducible". The second needs the authors' n, version and framing; the README states none of the three (we found no sample size; it names laya 0.3.20 where we run 0.3.28; prompts unknown). The methodology ruling (data-science 8df66c) therefore keeps three statements apart:

1. The criterion failed and is recorded as failed.
2. The harness is cleared for PAIRED Themis-vs-Laya use, on evidence, not on a story: the 100 pilot items inside the full run have the same text, options, gold and the same prediction 100/100 (95/100 correct); per-class accuracy is business 0.884, sci/tech 0.907, sports 0.967, world 0.935, top confusion business/sci-tech (169 + 126), so no degenerate class and no label-index defect; 0 runtime errors; routing english 7,598 / multilingual 2. These rule out the defects that would bias a paired comparison.
3. Leading explanation, not verified: the authors' 0.950 is a pilot-sized estimate. If the true accuracy is 0.9236, P(>= 95/100 on 100 items) = 0.215 - an unremarkable high draw - while the same 0.950 would be implausible at n >= 500 (P = 0.012) or n = 1,000 (P = 0.0006). The explanation holds only if the authors' n was small, which the README does not state. The version alternative (0.3.20 vs 0.3.28) was tested by a confirmatory run of laya 0.3.20 CODE on our pinned checkpoints, same 7,600 items, CPU fp32, offline (0.3.20 ships no pinned revisions, so "0.3.20 at its own checkpoints" is not a definable condition and was not run). Registered reading, declared before the result: point estimate inside [0.9174, 0.9293] or below = version EXCLUDED; its CI contains 0.950 = version EXPLAINS the gap; point estimate

above 0.9293 with CI not containing 0.950 = INCONCLUSIVE. Result: 7,019/7,600 = 0.9236 [0.9174, 0.9293], 0 failures, ECE 0.0563 - and the SAME prediction as 0.3.28 on every one of the 7,600 items, max |dprob| = 0.0. **Code version EXCLUDED** - at our checkpoints only: a different CHECKPOINT revision on the authors' side is not tested (0.3.20 has no pins, so that condition cannot be defined), and n is unknown. Sample size (with framing) stays the leading explanation, unverified.

"Laya at its best" on AG News is the default routing: 0.9236 against 0.9089 forced multilingual on the same 7,600 items (0 failures either way). The two routings differ in calibration as well as accuracy: default is under-confident (mean confidence 0.867 at 0.924 accuracy, ECE 0.056), multilingual better calibrated (0.939 at 0.909, ECE 0.032) - a second axis on which Themis is scored against Laya (THEMIS-RE-DESIGN W3).

Reporting rule from here (agreed with governance 0fa1fb as a publication condition): our comparator is "Laya 0.3.28, full split 0.924 [0.917, 0.929] (ours)"; 0.950 is the authors' unpaired figure, cited as such and never as a head-to-head; every Themis-vs-Laya number uses our measured Laya on identical items. The lesson is the one the pre-registration anticipated: n = 100 put the published figure inside a +/-5-point interval and we called it a match; 7,600 items did not.

4. What this fixes in the Themis comparison

- **Same task, free architecture.** Themis and Laya see identical items, identical option text and framing, and the same scoring and coverage rule; each system's budget and framing are recorded in the results file. Laya's budget is Laya's architecture, so Themis is not required to share it.
- **Laya at its best.** The comparator runs at its better configuration on DEV (default routing vs forced multilingual), recorded with revision and routing.
- **Design target W1.** Themis scores options with full context (option-wise or hierarchical), so that neither the option count nor the option wording can erase the distinctions between options.

5. Limitations

- The pre-declared AG News criterion failed on the full split (section 3a); the clearing of the harness is a decision under the pre-declared rule, disclosed as such, not a re-scored match. The confirmatory 0.3.20 run is reported in section 3a (version excluded); the forced-multilingual full-split leg is reported above (produced by the previous harness script bytes e739e0b9, as disclosed by ml-engineer 6e9bca; the data pin and sha gate added afterwards did not change the input bytes).
- n = 100 intervals are wide (about +/-10 points at 0.4); the Banking77 "consistent" is a weak screen by design.
- The default-routing rows at a 1,024 budget errored (100 failures) rather than scoring; reported as errors, not zeros.
- We found no stated sample size for Laya's published numbers at the pin, and their Banking77 used 77 labels where BTZSC uses 72; samples and prompts differ, which is why the match test is an interval check, not an equality.
- The wording finding rests on exploratory rows (one run each, n = 100, one benchmark) and on a contrast that changes wording and label set together; the paired tests make it decisive on these items, but it is a hypothesis for Themis's design, not a measured cause in general.
- The prefix-strip rows were ineffective (one word removed) and are kept only as a record; the shared-prefix hypothesis is untested here, and the option-0 collapse does not separate position from opening salience. The budget effect on the default routing is unmeasured (errors at 1,024).

Reproducibility

Table 4: Files.

Item	Where
Report files	`experiments/results/themis/laya-bench/` (sha256 9b538e40, 4868226e, ca7e680a, da3a3ade, 1ada1abd; git-ignored, rendered in /themis)
Harness	`scripts/judge/laya_bench.py`, `scripts/judge/laya_bench_b77.py` (ml-engineer; registration with the full-slice result)
Numbers in this note	`experiments/reports/_themis_bench001.py` prints and asserts each one
Pre-registration	ledger 7fd27f / 1c99a3 / 3ac327 / 41f8c6; Themis v1 plan `data/eval/experiments/PH-JUDGE-V1-PLAN.md`

Acknowledgments

We thank the leadership of Kooya PH for supporting this work, and the native-speaker evaluators who reviewed and rated the Filipino and Taglish material.

AI assistance. Analysis code, statistical checks and drafts of this document were prepared with the help of Claude (Anthropic), an AI model. Every number is generated from the project's result files. The author reviewed the content and takes responsibility for it.